Notes: Evans & Schwab (QJE, 1995)
Finishing High School and Starting College: Do Catholic Schools Make a Difference?

Contents

1 Big picture

- **Question.** Do Catholic high schools raise educational attainment, measured by high school completion and four-year college entry?
- **Why this matters.** The earlier Coleman debate focused heavily on standardized test scores. Evans and Schwab argue that graduating high school and starting college are more economically meaningful outcomes.
- **Main data.** The paper uses the High School and Beyond sophomore cohort: students who were sophomores in spring 1980 and were followed through 1984.
- **Main outcomes.** The two dependent variables are: (i) high school graduation by February 1984 and (ii) entry into a four-year college by February 1984, conditional on high school graduation.
- **Raw result.** In the raw data, Catholic school students graduate at a much higher rate than public school students, and they are more likely to enter four-year college.
- **Single-equation result.** Probit estimates imply that attending a Catholic high school raises the probability of graduating by about 12 to 13 percentage points and raises the probability of entering a four-year college by about 13 to 14 percentage points.
- **Robustness result.** These effects survive rich controls for family background, race, sex, family structure, religion, urban/rural status, region, test scores, peer composition, home educational resources, and state fixed effects.
- **Selection concern.** Catholic school attendance is chosen by families and possibly by schools. A high-attainment Catholic school effect could therefore reflect unobserved family motivation, student ability, or Catholic school screening rather than school effectiveness.
- **Identification strategy.** The paper uses bivariate probit models and instruments such as Catholic religion and percent Catholic in the county to test for selection bias.
- **Main selection result.** The authors find little evidence that the single-equation estimates are driven by selection bias. The estimated correlation between unobserved school-choice determinants and unobserved attainment determinants is small and statistically insignificant in most models.
- **Heterogeneity.** Catholic school effects are especially large for students with lower baseline probabilities of graduating, including black students, urban students, and students in the lowest test-score quartile. Effects remain positive for white and suburban students.
- **Takeaway.** The paper argues that Catholic high schools substantially increase educational attainment, not just test scores, and that the estimated effect is not obviously explained by selection.

2 Data and measurement

2.1 Data source: High School and Beyond

- The main data set is High School and Beyond (HS&B), a national survey of high school sophomores.
- Students were surveyed in 1980 and followed up in 1982 and 1984.
- HS&B includes student surveys, parent background variables, school questionnaires, and sophomore test scores.
- The analysis distinguishes public high school students and Catholic high school students.
- Table I reports 10,767 public school observations and 2,527 Catholic school observations.

Interpretation: HS&B is useful because it measures the student's high school sector before the outcomes occur and then follows the same student into high school completion and postsecondary enrollment decisions.

2.2 Outcome 1: high school graduate

The first dependent variable is

$$\text{HighSchoolGraduate}_i = \mathbf{1}\{\text{student } i \text{ graduated from high school by February 1984}\}.$$

- The raw graduation rate is 0.97 for Catholic school students and 0.79 for public school students.
- The main definition excludes GED recipients and students receiving diplomas after the February 1984 cutoff.
- The paper later checks more inclusive graduation definitions and finds a smaller but still economically meaningful Catholic school effect.

Interpretation: The authors focus on regular high school completion because it is a meaningful transition with important labor market consequences.

2.3 Outcome 2: four-year college entrant

The second dependent variable is

$$\text{CollegeEntrant}_i = \mathbf{1}\{\text{student } i \text{ first postsecondary school was a four-year college}\}.$$

- The college outcome is estimated conditional on having graduated from high school.
- In the raw data, the conditional four-year college entry rate is 0.55 for Catholic school students and 0.32 for public school students.
- The paper also checks broader definitions that include two-year colleges and later college entry.

Interpretation: Conditioning on high school graduation separates the college-entry margin from the graduation margin. Catholic schools may affect both.

2.4 Treatment variable: Catholic high school

The key treatment is

$$\text{CatholicSchool}_i = \mathbf{1}\{\text{student } i \text{ attends a Catholic high school}\}.$$

- Catholic school attendance is not randomly assigned.
- Families choose schools, and Catholic schools may screen students.
- The econometric challenge is to distinguish school quality from selection into Catholic schools.

Interpretation: The paper first estimates standard probit models and then tests for selection bias using bivariate probit models.

2.5 Controls

The baseline controls include:

- sex;
- race and ethnicity;
- family income categories;

- parents' education categories;
- family structure;
- student age;
- religious attendance;
- urban and rural school indicators;
- regional indicators.

Robustness controls include:

- sophomore test score and missing-test-score indicator;
- school peer-group composition by parental education and family income;
- home educational resources such as calculator, encyclopedia, books, and typewriter;
- state fixed effects recovered from the HS&B local labor market supplement.

Interpretation: The control strategy tries to remove observable differences between Catholic and public school students. The strongest threat remains unobserved selection.

2.6 Instrument variables used in selection models

The paper uses three types of instruments for Catholic school attendance:

- **Catholic religion:** a dummy equal to one if the student reports being Catholic.
- **Percent Catholic in the county:** a measure of local Catholic population share.
- **Interactions:** Catholic religion interacted with religious attendance or local Catholic share.

Interpretation: The instruments are intended to shift the cost or convenience of Catholic school attendance without directly shifting educational attainment, conditional on controls. The validity of this assumption is the central identifying issue in the bivariate probit section.

3 Models and empirical specifications

3.1 Single-equation latent-index model

For each educational outcome, the paper uses a latent variable model:

$$Y_i^* = X_i\beta + \gamma C_i + \varepsilon_i, \tag{1}$$

where

- Y_i^* is latent educational attainment;
- $Y_i = 1$ if $Y_i^* > 0$;
- C_i is Catholic school attendance;
- X_i is the vector of controls;
- γ measures the Catholic school effect in the probit index.

The observed outcome is

$$Y_i = \mathbf{1}\{Y_i^* > 0\}. \tag{2}$$

Interpretation: The single-equation model assumes Catholic school attendance is exogenous after conditioning on observables. If unobserved parental motivation or student ability affects both school choice and attainment, C_i is endogenous.

3.2 Marginal effect and average treatment effect

The paper reports both a marginal effect for a reference individual and an average treatment effect (ATE):

$$ATE = \frac{1}{N}\sum_{i=1}^{N}\left[\Phi(X_i\widehat{\beta}+\widehat{\gamma}) - \Phi(X_i\widehat{\beta})\right]. \tag{3}$$

Interpretation: The marginal effect answers: how much would Catholic school change the outcome probability for a specific benchmark student? The ATE averages this change across the sample's covariates.

3.3 Bivariate probit selection model

The paper then models Catholic school attendance as a second latent equation:

$$C_i^* = Z_i\delta + \mu_i, \tag{4}$$

with

$$C_i = \mathbf{1}\{C_i^* > 0\}. \tag{5}$$

The outcome and school-choice errors are assumed bivariate normal:

$$\begin{pmatrix}\varepsilon_i\\ \mu_i\end{pmatrix} \sim N\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & \rho\\ \rho & 1\end{pmatrix}\right). \tag{6}$$

Interpretation: If $\rho \neq 0$, unobserved determinants of school choice are correlated with unobserved determinants of educational attainment. A positive ρ would suggest positive selection into Catholic schools. A zero ρ supports the single-equation probit results.

3.4 Identification condition

The bivariate probit is identified when at least one variable in Z_i is excluded from X_i.

$$Z_i = (X_i, \text{instrument}_i).$$

Interpretation: Catholic religion and local Catholic share are used to create variation in Catholic school attendance. The identifying assumption is that these instruments do not directly affect graduation or college entry after controls.

3.5 Linear probability / 2SLS comparison

The paper also reports two-stage least squares estimates from linear probability models using the same instrument logic.

Interpretation: The 2SLS results are a robustness check against distributional assumptions in the bivariate probit. They generally point in a similar direction, though some instrument sets are much less precise.

4 Identification and empirical design

4.1 The core identification problem

Catholic school students differ from public school students in many ways:

- they are more likely to be Catholic;
- they are more likely to attend religious services;
- they come from somewhat more stable and educated family backgrounds;
- they may attend schools with more advantaged peers;
- they may be selected by the school through admissions or expulsion.

Interpretation: The main worry is that Catholic school effects are really family, student, or peer effects.

4.2 Omitted observable variables

The first line of defense is to add controls that could plausibly explain the Catholic school effect:

- test scores;
- missing-test-score indicators;
- peer-group composition;
- home educational resources;
- state fixed effects;
- all robustness variables together.

Interpretation: Table IV shows that even a very rich observable control set leaves an effect of roughly ten percentage points or more.

4.3 School selectivity

Catholic schools could select better students. The paper checks this by asking whether Catholic schools with entrance exams or waiting lists have stronger effects than Catholic schools without them.

Interpretation: Table V does not find such a pattern. The log-likelihood tests do not reject equality across more selective and less selective Catholic schools.

4.4 Unobserved selection

The second line of defense is the bivariate probit model. If unobserved parental involvement or student motivation drives both Catholic school choice and attainment, the estimated ρ should be important.

Interpretation: Table VI finds that ρ is small and statistically insignificant in most models. This is the paper's strongest evidence against selection bias.

4.5 Instrument validity

The paper studies alternative instruments and overidentification tests.

- Catholic religion is precise and yields stable results.
- Percent Catholic in county is less precise and performs poorly in some college-entry specifications.
- Combining instruments sometimes fails overidentification tests, especially in college-entry models.

Interpretation: The high school graduation results are more robust than the college-entry results. The paper's main conclusion is strongest for high school completion.

4.6 What the paper cannot fully prove

- Catholic school attendance is not randomized.
- Catholic religion may have direct effects through family culture or values.
- Local Catholic share may proxy for neighborhood or community characteristics.
- Peer effects may be endogenous and imperfectly controlled.
- The college outcome is conditional on graduation, so it is affected by sample selection into the graduate subsample.
- The paper predates modern randomized voucher evaluations and modern causal-inference designs.

Interpretation: The paper is careful to test many threats, but the causal interpretation still rests on maintained assumptions about instruments and unobserved selection.

5 Section-by-section detailed notes

5.1 Section I: Introduction

- The introduction starts from the Coleman debate over public versus Catholic school quality.
- The authors argue that prior work emphasized test scores, which may be less economically meaningful than actual educational attainment.
- They motivate high school graduation and college entry using evidence on unemployment, earnings, and returns to schooling.
- They preview the main result: Catholic high schools increase graduation and four-year college entry by approximately thirteen percentage points.
- They state the selection problem and preview bivariate probit models.

Economic message: The paper reframes school effectiveness away from cognitive-test gains and toward discrete life transitions with large economic consequences.

5.2 Section II: Data

- This section describes HS&B, the sophomore cohort, follow-up data, and the construction of outcomes.
- Table I shows that Catholic school students differ from public school students in religion, family structure, parental education, religious attendance, and test scores.

- The raw Catholic-public differences in graduation and college entry are large.
- Table II shows that the Catholic advantage appears across test-score quartiles, parental education groups, income groups, gender, and race.

Economic message: The data section establishes both the treatment contrast and the selection problem: Catholic students have better outcomes, but they also have different covariates.

5.3 Section III.A: Single-equation probit models

- Table III estimates probits for high school graduation and college entry.
- The Catholic school marginal effect is 0.117 for high school graduation and 0.144 for college entry.
- The average treatment effects are 0.130 and 0.132.
- The magnitude is large relative to other family background effects.
- The controls behave as expected: lower family income, lower parental education, and nontraditional family structure lower attainment.

Economic message: Conditional on observed background variables, Catholic school attendance remains a strong predictor of educational attainment.

5.4 Section III.B: Potential omitted variables bias

- The authors add sophomore test scores. This reduces the Catholic effect, but it remains large.
- Because many test scores are missing, they also impute missing test scores and include a missing indicator. This brings the Catholic effect close to the baseline.
- Adding peer measures does not eliminate the effect.
- Adding home educational resources reduces the effect somewhat but does not eliminate it.
- Adding state fixed effects leaves the result intact.
- Adding all robustness variables still leaves effects above ten percentage points.

Economic message: The estimates do not look like they are driven by a single omitted observable control.

5.5 Section III.C: Catholic school selectivity

- Catholic schools may screen students before admission or expel weak students.
- The paper uses entrance exams and waiting lists as proxies for school selectivity.
- If selection explains the Catholic effect, more selective Catholic schools should have larger effects.
- Table V finds no strong difference between Catholic schools with and without entrance exams or waiting lists.

Economic message: Direct proxies for school selectivity do not explain the main Catholic school effect.

5.6 Section III.D: Alternative dependent variables

- The authors broaden the graduation definition to include GEDs and late diplomas.
- The graduation effect falls to around eight percentage points, but remains positive.
- Broadening college entry to include two-year colleges and later enrollment leaves the Catholic school effect around twelve percentage points.

Economic message: The result is not purely an artifact of a narrow outcome definition.

5.7 Section IV.A: Bivariate probit model

- The paper treats Catholic school attendance as endogenous.
- Catholic religion is the initial instrument.
- Table VI shows that the bivariate probit estimates are close to single-equation estimates.
- The estimated error correlation ρ is not statistically significant.
- Adding test scores and state effects does not change the high school graduation conclusion, though the college-entry estimates fall in some specifications.

Economic message: The bivariate probit results provide little evidence that positive selection explains the high school graduation effect.

5.8 Section IV.B: Validity of the instruments

- The paper uses alternative instruments to test robustness.
- Catholic religion interacted with religious attendance gives similar high school graduation estimates.
- Percent Catholic in county is less precise and problematic for college entry.
- Overidentification tests are generally more favorable for high school graduation than for college entry.

Economic message: The instrument evidence strengthens the graduation result but leaves more ambiguity for college entry.

5.9 Section IV.C: Heterogeneity in Catholic school effects

- Table VIII studies high school graduation effects by race, urban/suburban location, test-score quartile, and religion.
- Effects are large for black students, urban students, and low-test-score students.
- Effects remain positive for white and suburban students.
- The effect for students in the highest test-score quartile is small, which makes sense because their public-school graduation rate is already high.

Economic message: Catholic schools appear most important for students at greater risk of not completing high school.

5.10 Section V: Summary and conclusions

- The paper concludes that Catholic high schools increase educational attainment.
- The authors emphasize that their results are robust to many observable controls and to models of selection.
- They also emphasize that the economic significance of graduation and college entry makes these outcomes more important than small test-score gains.

Economic message: The final message is that Catholic schools may matter not because they raise test scores substantially, but because they keep students on the path to high school completion and college entry.

6 Tables and original extracts

6.1 Table I: Summary statistics for HS&B

Read:

- Catholic school students have much higher raw graduation and college-entry rates: 0.97 versus 0.79 for high school graduation and 0.55 versus 0.32 for four-year college entry.
- Catholic school students are much more likely to be Catholic: 0.79 versus 0.29.
- Catholic students are more likely to live with both natural parents: 0.76 versus 0.62.
- Catholic students are more likely to attend religious services regularly: 0.69 versus 0.44.
- Catholic students have higher tenth-grade test scores: 30.06 versus 24.53.

Economic message: Table I shows both the Catholic school advantage and the selection problem. Catholic school students look different before the outcome is measured.

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TABLE I
SUMMARY STATISTICS: HIGH SCHOOL AND BEYOND DATA SET

Variable name	Definition	Catholic school mean and (std. dev.)	Public school mean and (std. dev.)
HIGH SCHOOL GRADUATE	0-1 dummy variable, = 1 if student graduated from high school by February of 1984	0.97 (0.17)	0.79 (0.41)
COLLEGE ENTRANT	0-1 dummy variable, = 1 if first postsecondary school attended was 4-year college	0.55[a] (0.50)	0.32[a] (0.47)
CATHOLIC RELIGION	0-1 dummy variable = 1 if the student is Catholic	0.79 (0.41)	0.29 (0.45)
% CATHOLIC IN COUNTY	Percent of the population in the county where the student attends school that is Catholic	31.65 (13.17)	22.37 (16.82)
FEMALE	0-1 dummy variable, = 1 if student is female	0.56 (0.50)	0.50 (0.50)
BLACK	0-1 dummy variable, = 1 if student is black	0.15 (0.36)	0.13 (0.34)
HISPANIC	0-1 dummy variable, = 1 if student is Hispanic	0.22 (0.41)	0.22 (0.41)
WHITE	0-1 dummy variable, = 1 is student is white, non-Hispanic	0.61 (0.49)	0.58 (0.49)
OTHER RACE	0-1 dummy variable, = 1 if student is other race	0.02 (0.15)	0.06 (0.24)
FAMILY INCOME MISSING	0-1 dummy variable, = 1 if family income is not reported	0.22 (0.41)	0.23 (0.42)
FAMILY INCOME < $7000	0-1 dummy variable, = 1 if family income < $7000	0.03 (0.16)	0.07 (0.26)
FAMILY INCOME $7000–$12,000	0-1 dummy variable, = 1 if family income ≥ $7000 and < $12,000	0.07 (0.26)	0.11 (0.31)
FAMILY INCOME $12,000–$16,000	0-1 dummy variable, = 1 if family income ≥ $12,000 and < $16,000	0.12 (0.32)	0.15 (0.35)
FAMILY INCOME $16,000–$20,000	0-1 dummy variable, = 1 if family income ≥ $16,000 and < $20,000	0.14 (0.35)	0.15 (0.35)
FAMILY INCOME $20,000–$25,000	0-1 dummy variable, = 1 if family income ≥ $20,000 and < $25,000	0.16 (0.36)	0.13 (0.33)
FAMILY INCOME $25,000–$38,000	0-1 dummy variable, = 1 if family income ≥ $25,000 and < $38,000	0.13 (0.33)	0.09 (0.29)
FAMILY INCOME ≥ $38,000	0-1 dummy variable, = 1 if family income ≥ $38,000	0.14 (0.35)	0.07 (0.25)
PARENT EDUCATION MISSING	0-1 dummy variable, = 1 if parents' education not reported	0.09 (0.29)	0.19 (0.40)
PARENT HIGH SCHOOL DROPOUT	0-1 dummy variable, = 1 if parents' highest education < high school graduate	0.23 (0.42)	0.30 (0.46)
PARENT HIGH SCHOOL GRADUATE	0-1 dummy variable, = 1 if parents' highest education is high school graduate	0.19 (0.39)	0.20 (0.40)
PARENT SOME COLLEGE	0-1 dummy variable, = 1 if parent's highest education is some college	0.28 (0.45)	0.19 (0.39)

Table I, part 1

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TABLE I
(CONTINUED)

Variable name	Definition	Catholic school mean and (std. dev.)	Public school mean and (std. dev.)
PARENT COLLEGE GRADUATE	0-1 dummy variable, = 1 if parents' highest education is college graduate	0.21 (0.41)	0.11 (0.31)
SINGLE MOTHER	0-1 dummy variable, = 1 if student's household is headed by single mother	0.12 (0.32)	0.15 (0.35)
SINGLE FATHER	0-1 dummy variable, = 1 if student's household is headed by single father	0.03 (0.17)	0.05 (0.21)
NATURAL MOTHER/ STEPFATHER	0-1 dummy variable, = 1 if student lives with natural mother and stepfather	0.04 (0.19)	0.06 (0.24)
BOTH NATURAL PARENTS	0-1 dummy variable, = 1 if student lives with both natural parents	0.76 (0.43)	0.62 (0.48)
OTHER FAMILY STRUCTURE	0-1 dummy variable, = 1 if student's household has other structure	0.06 (0.24)	0.12 (0.32)
AGE 16	0-1 dummy variable, = 1 if student is <=16 years of age in February of 1982	0.03 (0.17)	0.03 (0.17)
AGE 17	0-1 dummy variable, = 1 if student is 17 years of age in February of 1982	0.63 (0.48)	0.49 (0.50)
AGE 18	0-1 dummy variable, = 1 if student is 18 years of age in February of 1982	0.32 (0.47)	0.40 (0.49)
AGE 19+	0-1 dummy variable, = 1 is student is 19 years of age or older	0.02 (0.15)	0.08 (0.26)
ATTENDS RELIGIOUS SERVICES REGULARLY	0-1 dummy variable, = 1 if student attends church at least twice a month	0.69 (0.46)	0.44 (0.50)
ATTENDS RELIGIOUS SERVICES OCCASIONALLY	0-1 dummy variable, = 1 if student attends church occasionally	0.17 (0.38)	0.23 (0.42)
NEVER ATTENDS RELIGIOUS SERVICES	0-1 dummy variable, = 1 if student never attends church	0.13 (0.34)	0.33 (0.47)
10th GRADE TEST SCORE	Student's sophomore score on standardized exam	30.06 (14.63)	24.53 (15.87)
TEST SCORE MISSING	0-1 dummy variable, = 1 if sophomore test score is missing	0.08 (0.28)	0.16 (0.37)
No. of obs.		10,767	2527

Table I, part 2

6.2 Table II: Raw educational outcomes by school type

Read:

- In the full sample, Catholic students graduate at 0.97 versus 0.79 for public students.
- Conditional on graduation, Catholic students enter four-year college at 0.55 versus 0.32 for public students.
- The Catholic advantage appears in every sophomore test-score quartile.
- The Catholic advantage also appears across parental education, family income, gender, and racial groups.
- Among black students, the college-entry rate is 0.62 for Catholic school students versus 0.33 for public school students.

Economic message: The raw Catholic school advantage is not confined to one demographic subgroup.

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TABLE II
EDUCATIONAL OUTCOMES OF HIGH SCHOOL STUDENTS BY SCHOOL TYPE

	HIGH SCHOOL GRADUATE		*COLLEGE ENTRANT*[a]	
Sample	Public schools	Catholic schools	Public schools	Catholic schools
Full sample	0.79	0.97	0.32	0.55
SOPHOMORE TEST SCORE MISSING	0.71	0.98	0.22	0.50
SOPHOMORE TEST FIRST QUARTILE	0.63	0.91	0.11	0.25
SOPHOMORE TEST SECOND QUARTILE	0.80	0.96	0.19	0.40
SOPHOMORE TEST THIRD QUARTILE	0.89	0.98	0.37	0.56
SOPHOMORE TEST FOURTH QUARTILE	0.95	0.99	0.62	0.78
PARENT EDUCATION MISSING	0.65	0.92	0.16	0.40
PARENT H.S. DROPOUT	0.77	0.95	0.22	0.41
PARENT H.S. DEGREE	0.82	0.97	0.30	0.54
PARENT SOME COLLEGE	0.87	0.98	0.44	0.62
PARENT COLLEGE GRADUATE	0.91	0.99	0.61	0.67
FAMILY INCOME MISSING	0.74	0.97	0.25	0.48
FAMILY INCOME < $7000	0.64	0.91	0.19	0.36
FAMILY INCOME $7000–$12000	0.76	0.92	0.23	0.44
FAMILY INCOME $12000–$16000	0.81	0.98	0.29	0.51
FAMILY INCOME $16000–$20000	0.84	0.97	0.33	0.49
FAMILY INCOME $20000–$25000	0.84	0.96	0.38	0.57
FAMILY INCOME $25000–$38000	0.87	0.99	0.47	0.70
FAMILY INCOME > $38000	0.86	0.98	0.52	0.66

6.3 Table III: Baseline probit models

Read:

- Catholic school coefficient in the graduation probit is 0.777, with marginal effect 0.117.
- Catholic school coefficient in the college-entry probit is 0.384, with marginal effect 0.144.
- The average treatment effects are 0.130 for graduation and 0.132 for college entry.
- Parental education, income, family structure, and religious attendance are important controls.
- The Catholic school effect is large relative to most family background variables.

Economic message: Conditional on observables, Catholic school attendance predicts a 12 to 14 percentage point increase in attainment.

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TABLE III
PROBIT ESTIMATES OF *HIGH SCHOOL GRADUATE* AND *COLLEGE ENTRANT* MODELS

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TABLE III (CONTINUED)

	HIGH SCHOOL GRADUATE		*COLLEGE ENTRANT*	
Independent variable[a]	Probit coefficient	Marginal effect[b]	Probit coefficient	Marginal effect[b]
CATHOLIC SCHOOL	0.777	0.117	0.384	0.144
	(0.056)	(0.014)	(0.032)	(0.012)
FEMALE	0.041	0.006	0.021	0.008
	(0.029)	(0.004)	(0.026)	(0.010)
BLACK	0.132	0.020	0.170	0.064
	(0.045)	(0.007)	(0.042)	(0.014)
HISPANIC	0.080	0.012	−0.160	−0.060
	(0.037)	(0.006)	(0.036)	(0.014)
OTHER RACE	0.346	0.052	0.316	0.118
	(0.067)	(0.011)	(0.060)	(0.022)
FAMILY INCOME MISSING	−0.111	−0.017	−0.382	−0.143
	(0.068)	(0.010)	(0.055)	(0.021)
FAMILY INCOME < $7000	−0.300	−0.045	−0.484	−0.181
	(0.078)	(0.012)	(0.080)	(0.030)
FAMILY INCOME	−0.121	−0.018	−0.408	−0.153
$7000–$12,000	(0.073)	(0.011)	(0.063)	(0.024)
FAMILY INCOME	−0.035	−0.005	−0.319	−0.119
$12,000–$16,000	(0.072)	(0.011)	(0.056)	(0.021)
FAMILY INCOME	0.000	0.000	−0.283	−0.106
$16,000–$20,000	(0.070)	(0.010)	(0.055)	(0.020)
FAMILY INCOME	−0.035	−0.005	−0.196	−0.073
$20,000–$25,000	(0.072)	(0.011)	(0.055)	(0.021)
FAMILY INCOME	0.037	0.006	−0.025	−0.009
$25,000–$38,000	(0.077)	(0.012)	(0.057)	(0.021)
PARENT EDUCATION	−0.730	−0.110	−0.916	−0.342
MISSING	(0.061)	(0.013)	(0.052)	(0.020)
PARENT HIGH SCHOOL	−0.522	−0.078	−0.855	−0.320
DROPOUT	(0.058)	(0.011)	(0.043)	(0.017)
PARENT HIGH SCHOOL	−0.375	−0.056	−0.602	−0.225
GRADUATE	(0.060)	(0.011)	(0.044)	(0.015)
PARENT SOME COLLEGE	−0.204	−0.031	−0.290	−0.108
	(0.062)	(0.010)	(0.042)	(0.016)
SINGLE MOTHER	−0.255	−0.038	−0.060	−0.023
	(0.041)	(0.007)	(0.042)	(0.016)
SINGLE FATHER	−0.421	−0.063	−0.269	−0.101
	(0.063)	(0.010)	(0.069)	(0.026)
NATURAL MOTHER/STEP-	−0.286	−0.043	−0.263	−0.098
FATHER	(0.056)	(0.009)	(0.060)	(0.023)
OTHER FAMILY STRUC-	−0.155	−0.023	−0.060	−0.023
TURE	(0.048)	(0.007)	(0.053)	(0.020)
AGE 16	0.611	0.092	0.655	0.245
	(0.089)	(0.015)	(0.115)	(0.043)
AGE 17	1.025	0.154	0.718	0.268
	(0.050)	(0.014)	(0.087)	(0.033)
AGE 18	0.699	0.105	0.603	0.225
	(0.050)	(0.012)	(0.088)	(0.033)
ATTENDS RELIGIOUS SER-	0.321	0.048	0.299	0.112
VICES REGULARLY	(0.035)	(0.006)	(0.035)	(0.014)
ATTEND RELIGIOUS SER-	0.082	0.012	0.115	0.043
VICES OCCASIONALLY	(0.039)	(0.006)	(0.041)	(0.015)
INTERCEPT	0.388		−0.683	
	(0.093)		(0.107)	
Average treatment effect of	0.130		0.132	
CATHOLIC SCHOOL	(0.007)		(0.011)	
Log/Likelihood	−5155.26		−3297.87	

Asymptotic standard errors are in parentheses. The number of observations in the *HIGH SCHOOL GRADUATE* and *COLLEGE ENTRANT* models is 13,294 and 10,983, respectively.
a. Other exogenous variables include dummy variables for urban and rural schools, plus three regional dummy variables.
b. Marginal effects are calculated for a seventeen-year old white female, living with both natural parents where at least one parent has a high school degree and family income is between $16,000 and $20,000, attends

Table III, part 1 Table III, part 2

6.4 Table IV: Robustness to additional controls

Read:

- Adding tenth-grade test score reduces the graduation ATE from 0.130 to 0.100 and the college ATE from 0.132 to 0.111.
- Adding test score with a missing-score indicator restores effects close to baseline: 0.120 for graduation and 0.120 for college entry.
- Adding peer-group controls produces ATEs of 0.123 and 0.104.
- Adding home educational resources produces ATEs of 0.111 and 0.117, or 0.128 and 0.130 when missing indicators are included.
- Adding state effects yields ATEs of 0.136 and 0.140.
- Adding all of the major robustness controls still gives ATEs of 0.120 and 0.108.

Economic message: Observable omitted variables do not explain away the Catholic school effect. The fully controlled model still implies effects above ten percentage points.

TABLE IV

PROBIT ESTIMATES OF *HIGH SCHOOL GRADUATE* AND *COLLEGE ENTRANT* MODELS

		HIGH SCHOOL GRADUATE				*COLLEGE ENTRANT*			
Model	Additional exogenous variables[a]	No. of obs.	Probit coefficient on *CATHOLIC SCHOOL*	Marginal effect[b]	Average treatment effect	No. of obs.	Probit coefficient on *CATHOLIC SCHOOL*	Marginal effect[b]	Average treatment effect
(1)		13,294	0.777 (0.056)	0.117 (0.014)	0.130 (0.007)	10,983	0.384 (0.032)	0.144 (0.012)	0.132 (0.011)
(2)	*10th GRADE TEST SCORE*	11,379	0.632 (0.061)	0.093 (0.013)	0.100 (0.008)	9,567	0.367 (0.036)	0.125 (0.012)	0.111 (0.011)
(3)	*10th GRADE TEST SCORE* and a dummy variable for missing test score	13,294	0.722 (0.059)	0.112 (0.014)	0.120 (0.007)	10,983	0.392 (0.034)	0.135 (0.012)	0.120 (0.011)
(4)	Measures of peer groups[c]	13,294	0.726 (0.058)	0.129 (0.016)	0.123 (0.007)	10,983	0.308 (0.034)	0.107 (0.012)	0.104 (0.012)
(5)	Dummy variables for whether family owns a calculator, encyclopedia, more than 50 books, or a typewriter	10,797	0.710 (0.062)	0.097 (0.014)	0.111 (0.007)	9,266	0.337 (0.035)	0.130 (0.013)	0.117 (0.012)
(6)	Dummy variables for whether family owns a calculator, encyclopedia, more than 50 books, or a typewriter, and dummy variables for whether these values are missing	13,294	0.764 (0.057)	0.106 (0.013)	0.128 (0.007)	10,983	0.382 (0.032)	0.146 (0.012)	0.130 (0.012)
(7)	State dummy variables	13,294	0.826 (0.058)	0.132 (0.016)	0.136 (0.007)	10,983	0.415 (0.034)	0.134 (0.011)	0.140 (0.012)
(8)	All of the variables in models (3), (4), (6), and (7)	13,294	0.735 (0.061)	0.137 (0.019)	0.120 (0.007)	10,983	0.369 (0.037)	0.111 (0.011)	0.108 (0.012)

Asymptotic standard errors are in parentheses.

a. Other right-hand-side variables include those listed in Table III.

b. Marginal effects are calculated for the individual defined in Table III. The medium public school test scores and average public school values for the peer group measures are used in the appropriate models. For models (5), (6), and (8), individuals are assumed to own all items listed. The reference state used in models (7) and (8) is the state with the most observations in the sample.

c. A set of seven variables that measure the percent of students in a high school whose parents fall into four education categories and whose family falls into three income categories.

6.5 Table V: Catholic school selectivity proxies

Read:

- Catholic schools with entrance exams have graduation marginal effect 0.117.
- Catholic schools without entrance exams have graduation marginal effect 0.159.
- Catholic schools with waiting lists have graduation marginal effect 0.128.
- Catholic schools without waiting lists have graduation marginal effect 0.118.
- The log-likelihood tests do not reject equality across school types.

Economic message: The Catholic effect is not stronger in the schools that look more selective by entrance exams or waiting lists.

6.6 Table VI: Bivariate probit using Catholic religion as instrument

Read:

- Baseline single-equation estimates are repeated for comparison.
- Using Catholic religion as an instrument, the high school graduation ATE is 0.141.
- The college-entry ATE is 0.098 in the comparable bivariate probit.
- The estimated ρ is small and statistically insignificant in the main models.
- Adding test scores and state effects does not materially change the graduation result.
- College-entry estimates are less stable once test scores are added.

Economic message: The selection-corrected estimates support the high school graduation result and provide weaker but still positive evidence for college entry.

6.7 Table VII: Alternative instruments

Read:

- Catholic religion alone gives a graduation ATE of 0.141 and a 2SLS estimate of 0.127.
- Catholic religion interacted with religious attendance gives similar graduation effects.

TABLE V

PROBIT ESTIMATES OF *HIGH SCHOOL GRADUATE* AND *COLLEGE ENTRANT* MODELS

Independent variables[a]	% of Catholic school students	*HIGH SCHOOL GRADUATE* Probit coefficient	Marginal effect[b]	Probit coefficient	Marginal effect[b]	*COLLEGE ENTRANT* Probit coefficient	Marginal effect[b]	Probit coefficient	Marginal effect[b]
CATHOLIC SCHOOLS WITH ENTRANCE EXAMS	0.842	0.778	0.117			0.388	0.145		
		(0.064)	(0.014)			(0.036)	(0.013)		
CATHOLIC SCHOOLS WITHOUT ENTRANCE EXAMS	0.158	1.059	0.159			0.333	0.124		
		(0.178)	(0.030)			(0.072)	(0.027)		
CATHOLIC SCHOOLS WITH WAITING LISTS	0.512			0.850	0.128			0.378	0.141
				(0.087)	(0.018)			(0.044)	(0.016)
CATHOLIC SCHOOLS WITHOUT WAITING LISTS	0.488			0.782	0.118			0.379	0.141
				(0.081)	(0.016)			(0.044)	(0.016)
−2 loglikelihood test[c]		2.40		0.36		0.52		0.02	

Asymptotic standard errors are in parentheses. Answers to the entrance exam and waiting list questions were missing in some cases. The number of observations (mean of dependent variable) in the *HIGH SCHOOL GRADUATE* and *COLLEGE ENTRANT* models is 13,033 (0.821) and 10,735 (0.373), respectively.

a. Other right-hand-side variables include those listed in Table III.

b. The marginal effect is calculated for the individual described in Table III.

c. The test statistic is the statistic required to test the equality of the coefficients on the two types of Catholic schools. The test is asymptotically distributed as a χ^2 with one degree of freedom. The 95 percent critical value is 3.84.

TABLE VI

MAXIMUM LIKELIHOOD ESTIMATES OF *HIGH SCHOOL GRADUATE* AND *COLLEGE ENTRANT* BIVARIATE PROBIT MODEL USING *CATHOLIC RELIGION* AS AN INSTRUMENT

Model	Other variables in X_i[b]	MLE estimates of bivariate probit model: Coefficient on *CATHOLIC SCHOOL*	Marginal effect[c]	Average treatment effect	ρ	2SLS estimate of coefficient on *CATHOLIC SCHOOL*
HIGH SCHOOL GRADUATE[a]						
(1)		0.777	0.117	0.130		0.096[d]
		(0.056)	(0.014)	(0.007)		(0.008)
(2)		0.859	0.133	0.141	−0.053	0.127
		(0.115)	(0.022)	(0.014)	(0.067)	(0.024)
(3)	*10TH GRADE TEST SCORE AND TEST MISSING*	0.678	0.078	0.114	0.028	0.103
		(0.126)	(0.018)	(0.017)	(0.072)	(0.024)
(4)	*STATE EFFECTS*	0.911	0.142	0.144	−0.050	0.114
		(0.121)	(0.027)	(0.015)	(0.072)	(0.024)
(5)	*10TH GRADE TEST SCORE, TEST MISSING, AND STATE EFFECTS*	0.746	0.124	0.121	0.025	0.134
		(0.132)	(0.028)	(0.016)	(0.077)	(0.030)
COLLEGE ENTRANT[a]						
(6)		0.384	0.144	0.132		0.137[d]
		(0.032)	(0.012)	(0.011)		(0.011)
(7)		0.288	0.109	0.098	0.067	0.148
		(0.079)	(0.033)	(0.028)	(0.049)	(0.030
(8)	*10TH GRADE TEST SCORE AND TEST MISSING*	0.211	0.078	0.064	0.124	0.098
		(0.083)	(0.034)	(0.026)	(0.052)	(0.024)
(9)	*STATE EFFECTS*	0.341	0.110	0.115	0.056	0.092
		(0.084)	(0.032)	(0.029)	(0.053)	(0.024)
(10)	*10TH GRADE TEST SCORE, TEST MISSING, AND STATE EFFECTS*	0.277	0.071	0.082	0.113	0.098
		(0.090)	(0.026)	(0.027)	(0.046)	(0.028)

Asymptotic standard errors are in parentheses.

a. Models (1) and (6) are single-equation estimates from Table III. To estimate models (4), (5), (9), and (10), we deleted all states with no Catholic school students. The high school completion and college entrance models contain 10,120 and 8470 observations, respectively. Both models contain data from twenty states. Models (1), (2), and (3) contain 13,294 observations, and models (6), (7), and (8) contain 10,983 observations.

b. Other exogenous variables include those listed in Table III.

c. Marginal effects are calculated for the individual defined in Table III.

d. Estimated *CATHOLIC SCHOOL* coefficient from a linear probability model.

significantly different from zero. Even in these models, however, attending a Catholic high school increases the probability of entering college by more than seven percentage points.

- Percent Catholic in county gives a graduation ATE of 0.114 but with a larger standard error.
- For college entry, percent Catholic in county produces much larger estimates and fails overidentification tests when combined with Catholic religion.
- The most stable instrument for the main result is Catholic religion.

Economic message: The instrument evidence is credible for graduation but more fragile for college entry.

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TABLE VII
SYSTEM ESTIMATES OF *HIGH SCHOOL GRADUATE* AND *COLLEGE ENTRANT* MODELS WITH ALTERNATIVE INSTRUMENTS

Instruments	Bivariate probit estimates of average treatment effect, *CATHOLIC SCHOOL*	2SLS estimate of *CATHOLIC SCHOOL*	Test of overidentifying restrictions, (d.o.f.), [95% critical value]
HIGH SCHOOL GRADUATE[a]			
(1) *CATHOLIC RELIGION*	0.141 (0.014)	0.127 (0.024)	
(2) *CATHOLIC RELIGION × ATTENDANCE AT RELIGIOUS SERVICES*	0.141 (0.013)	0.107 (0.022)	3.29 (2) [5.99]
(3) *% CATHOLIC IN COUNTY*	0.114 (0.033)	0.130 (0.076)	
(4) *CATHOLIC RELIGION and % CATHOLIC IN COUNTY*	0.139 (0.044)	0.127 (0.024)	0.10 (1) [3.84]
(5) *CATHOLIC RELIGION, % CATHOLIC IN COUNTY AND CATHOLIC RELIGION* % CATHOLIC IN COUNTY*	0.137 (0.014)	0.127 (0.024)	0.84 (2) [5.99]
(6) *% CATHOLIC IN COUNTY*[b]	0.061 (0.038)	0.144 (0.373)	
COLLEGE ENTRANT[a]			
(7) *CATHOLIC RELIGION*	0.098 (0.028)	0.148 (0.030)	
(8) *CATHOLIC RELIGION × ATTENDANCE AT RELIGIOUS SERVICES*	0.122 (0.127)	0.167 (0.027)	6.3 (2) [5.99]
(9) *% CATHOLIC IN COUNTY*	0.240 (0.053)	0.656 (0.093)	
(10) *CATHOLIC RELIGION and % CATHOLIC IN COUNTY*	0.115 (0.037)	0.161 (0.029)	33.7 (1) [3.84]
(11) *CATHOLIC RELIGION and CATHOLIC RELIGION × % CATHOLIC IN COUNTY*[c]	0.071 (0.028)	0.104 (0.031)	0.81 (1) [3.84]

Asymptotic standard errors are in parentheses. The number of observations in the *HIGH SCHOOL GRADUATE* and *COLLEGE ENTRANT* models is 13,294 and 10,983, respectively.
a. Other exogenous variables include those listed in Table III.
b. *CATHOLIC RELIGION* is included as an exogenous variable in the model.
c. *% CATHOLIC IN COUNTY* is included as an exogenous variable in the model.

results are fairly robust, though the results where we depend on *CATHOLIC RELIGION* as an instrument are estimated more precisely.

The *COLLEGE ENTRANT* models in lines (7) through (10)

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TABLE VIII
HETEROGENEITY OF THE AVERAGE TREATMENT EFFECT, *HIGH SCHOOL GRADUATE* MODELS

			Average treatment effect, *CATHOLIC SCHOOL*[a]		
				Bivariate probit estimates with instructions:	
Sample	Number of obs.	Mean *HIGH SCHOOL GRADUATE*	Single-equation probit	*% CATHOLIC IN COUNTY*	*CATHOLIC RELIGION*
WHITE	7831	0.826	0.141 (0.007)	0.086 (0.039)	0.128 (0.016)
BLACK	1833	0.803	0.134 (0.019)	0.111 (0.101)	0.146 (0.044)
URBAN[b]	3150	0.774	0.172 (0.016)	0.139 (0.069)	0.184 (0.037)
SUBURBAN	6696	0.862	0.109 (0.008)	−0.003 (0.052)	0.120 (0.017)
SOPHOMORE TEST, FIRST QUARTILE	2842	0.658	0.213 (0.025)	0.113 (0.145)	0.242 (0.051)
SOPHOMORE TEST, SECOND QUARTILE	2842	0.829	0.105 (0.016)	0.128 (0.066)	0.110 (0.037)
SOPHOMORE TEST, THIRD QUARTILE	2854	0.916	0.069 (0.010)	0.176 (0.039)	0.071 (0.020)
SOPHOMORE TEST, FOURTH QUARTILE	2841	0.960	0.030 (0.007)	−0.217 (0.188)	0.012 (0.031)
CATHOLIC	5104	0.884	0.107 (0.008)	0.328 (0.033)	
NON-CATHOLIC	8190	0.790	0.145 (0.013)	0.072 (0.098)	

Asymptotic standard errors are in parentheses.
a. Other exogenous variables include those listed in Table III.
b. Schools in the South were deleted from this subsample because there were no urban Catholic schools.

6.8 Table VIII: Heterogeneity of high school graduation effect

Read:

- Single-equation ATEs are 0.141 for white students and 0.134 for black students.
- Urban students have a larger ATE, 0.172, than suburban students, 0.109.
- Students in the first test-score quartile have the largest single-equation ATE, 0.213.
- Effects fall as test-score quartile rises, which reflects higher baseline graduation rates among high scorers.
- The Catholic school effect is positive for both Catholic and non-Catholic students.

Economic message: Catholic schools appear most consequential for students with relatively low baseline graduation probabilities.

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TABLE VIII
HETEROGENEITY OF THE AVERAGE TREATMENT EFFECT, *HIGH SCHOOL GRADUATE* MODELS

			Average treatment effect, *CATHOLIC SCHOOL*[a]		
				Bivariate probit estimates with instructions:	
Sample	Number of obs.	Mean *HIGH SCHOOL GRADUATE*	Single-equation probit	*% CATHOLIC IN COUNTY*	*CATHOLIC RELIGION*
WHITE	7831	0.826	0.141 (0.007)	0.086 (0.039)	0.128 (0.016)
BLACK	1833	0.803	0.134 (0.019)	0.111 (0.101)	0.146 (0.044)
URBAN[b]	3150	0.774	0.172 (0.016)	0.139 (0.069)	0.184 (0.037)
SUBURBAN	6696	0.862	0.109 (0.008)	−0.003 (0.052)	0.120 (0.017)
SOPHOMORE TEST, FIRST QUARTILE	2842	0.658	0.213 (0.025)	0.113 (0.145)	0.242 (0.051)
SOPHOMORE TEST, SECOND QUARTILE	2842	0.829	0.105 (0.016)	0.128 (0.066)	0.110 (0.037)
SOPHOMORE TEST, THIRD QUARTILE	2854	0.916	0.069 (0.010)	0.176 (0.039)	0.071 (0.020)
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CATHOLIC	5104	0.884	0.107 (0.008)	0.328 (0.033)	
NON-CATHOLIC	8190	0.790	0.145 (0.013)	0.072 (0.098)	

Asymptotic standard errors are in parentheses.
a. Other exogenous variables include those listed in Table III.
b. Schools in the South were deleted from this subsample because there were no urban Catholic schools.

7 Short summary

- The paper studies whether Catholic high schools improve educational attainment.
- It uses HS&B sophomore cohort data and follows students to 1984.
- Outcomes are high school graduation and four-year college entry.
- Raw data show large Catholic-public differences in both outcomes.
- Baseline probit models estimate a Catholic school effect of roughly 12 to 14 percentage points.
- Robustness controls for test scores, peer groups, home resources, and state effects do not eliminate the effect.
- Entrance exams and waiting lists do not explain the effect.
- Bivariate probit models using Catholic religion as an instrument find little evidence of selection bias.
- Alternative instruments support the graduation result more strongly than the college-entry result.
- Heterogeneity results suggest larger gains for black, urban, and low-test-score students.
- The paper concludes that Catholic schools have economically important effects on educational attainment.

8 Conclusion

8.1 Core conclusion

Evans and Schwab argue that Catholic high schools significantly increase the probability that students complete high school and start four-year college. Their baseline estimates imply effects of about thirteen percentage points, and the graduation result is robust to many controls and to selection-correction models.

8.2 Economic intuition

The economic mechanism is not fully pinned down, but the results are consistent with Catholic schools improving persistence and educational continuation. This could reflect discipline, peer environment, parental-school networks, expectations, monitoring, or organizational structure. The paper's main empirical claim is that whatever the mechanism, the attainment effect is large and not easily explained by observed student background.

8.3 Takeaway

The paper shifts the public-versus-Catholic school debate from test scores to attainment. Its most durable lesson is that school quality should be judged by economically meaningful outcomes. For this sample, Catholic high school attendance is associated with substantially higher probabilities of finishing high school and entering four-year college.